Juggling Probabilities

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Gregory S. Warrington

1. INTRODUCTION. Imagine yourself effortlessly juggling five balls in a high, lazy pattern. Your right hand catches a ball and immediately throws it. One second later, it is your left hand's turn to catch and throw a ball. Then it is your right hand's turn again. . . . Some of your throws may be low, some high. Some balls go straight up; others cross over to the opposite hand. At most times, one ball lands; occasionally a hand remains empty. But the alternating cadence of your hands is unwavering.

Suppose that, while you are juggling, some singular force of nature causes time to stand still for a moment. Certain balls are in the air—you have already thrown them. To avoid dropping the ball, you must make catches at certain times in the future. If your previous few throws were all low, perhaps you are committed only at one, two, four, six, and seven seconds into the future. On the other hand, if you have just vigorously launched a ball, you might be committed at one, two, three, four, and ten seconds into the future. The set of "committed times" is your *landing state* (*state*, for short). As you juggle, you wander from state to state according to what throw you have made most recently.

Our goal in this paper is to answer the following question:

Random question. What fraction of the time is spent in any given state if every throw is made randomly?

The answer, of course, depends on how we specify a certain set of parameters:

1. the possible states,
2. the throws that are possible from a given state, and
3. the probability of making each legal throw.

There are countless ways to make these specifications, but we begin our investigations with the model that most closely mirrors what people envision as juggling. In the next section, we describe the juggling universe determined by this standard model and answer our random question. In section 3 we generalize the notion of a state in order to give a simple justification for our answer. Finally, in section 4 we describe variations of this model.

2. JUGGLING STATES. Jugglers have been known to

$$\{\text{toss, roll, spin, bounce, drop, kick}\}$$

their

$$\{\text{balls, rings, clubs, torches, chairs}\}$$

while accompanying their acts with

$$\{\text{patter, music, dance, flourishes}\}.$$

We, however, intend to strip juggling down to what is (arguably) its essentials. Our juggler will juggle only balls and only in the air. He will not attempt jokes. Whether a hand is under a leg or behind the back when making a catch will not be noted. *We record only the order in which the balls are thrown and caught.* While clearly ignoring much of what makes juggling visually intriguing, our single-mindedness will free us to highlight the inherently mathematical nature of juggling.

Our juggler will:

1. throw alternately from each hand, once a second;
2. throw a ball instantaneously upon catching it;
3. be assumed to have always been (and be forevermore) juggling (this allows us to avoid starting/stopping issues);
4. be named Magnus.[1]

Every ball Magnus throws lands some (integral) number of seconds into the future. By abuse of physics, we will refer to this parameter as the *height* of the throw. For the time being we assume the existence of some maximum height h above which Magnus is too weak to throw.

In the introduction, we referred to a state as a set of future times at which balls land. It is more convenient for the purposes of this paper to define a *(landing) state* $\boldsymbol{\nu}$ as a word of length h in the alphabet $\{\bullet, \smile\}$. Given a state $\boldsymbol{\nu}$, we write ν_t for its tth letter and express $\boldsymbol{\nu}$ as

$$\boldsymbol{\nu} = \boxed{\nu_1\, \nu_2 \cdots \nu_h}\,.$$

If $\nu_t = \bullet$, then Magnus will catch a ball t seconds into the future; if $\nu_t = \smile$, then no ball lands t seconds in the future. We make the convention that $\nu_t = \smile$ whenever $t > h$. To recover our original notion of a state from $\boldsymbol{\nu}$, we take the set $\{t : \nu_t = \bullet\}$. We denote the set of all h-letter states by St_h. The subset $St_{h,f}$ of St_h comprises those h-letter states in which $\smile$ occurs exactly f times.

As an example, consider the state $\boldsymbol{\nu} = \boxed{\bullet\,\bullet\,\smile\,\bullet\,\smile\,\bullet\,\bullet\,\bullet\,\smile\,\smile}$ in St_{10}. Assuming that Magnus has just thrown out of his left hand, at one and seven seconds of lapsed time he will be (instantaneously) catching/throwing with his right hand; at two, four, six and eight seconds into the future he will be (instantaneously) catching/throwing with his left hand.

When the first letter of a state $\boldsymbol{\nu} = \boxed{\nu_1\, \nu_2 \cdots \nu_h}$ is $\smile$, we often refer to what Magnus does during the next second as making a "height 0" throw. Upon making a throw (even one of height 0), Magnus commits himself to a (usually different) new state $\boldsymbol{\omega}$ in accordance with the following two rules:

1. if the throw is of height 0, then $\boldsymbol{\omega} = \boxed{\nu_2\, \nu_3 \cdots \nu_h\, \smile}$;
2. if the throw is of height $t (\geq 1)$, then $\boldsymbol{\omega} = \boxed{\nu_2\, \nu_3 \cdots \bullet \cdots \nu_h\, \smile}$, where the $\bullet$ is placed in the tth position in $\boldsymbol{\omega}$ (replacing ν_{t+1} in $\boldsymbol{\nu}$).

As an example, we illustrate successive throws of heights 2, 0, and 6 from an initial state in which balls land at one, four, and six seconds into the future:

$$\boxed{\bullet\,\smile\,\smile\,\bullet\,\smile\,\bullet} \xrightarrow{2} \boxed{\smile\,\bullet\,\bullet\,\smile\,\bullet\,\smile} \xrightarrow{0} \boxed{\bullet\,\bullet\,\smile\,\bullet\,\smile\,\smile} \xrightarrow{6} \boxed{\bullet\,\smile\,\bullet\,\smile\,\smile\,\bullet}\,.$$

In Figure 1, we show all possible states and transitions when Magnus has three balls and is not able to throw anything higher than a four.

[1] After Magnus Nicholls, a juggler prominent during the early 1900s. He is reputed to be the first person to juggle five clubs. Reliable details of his life are outclassed by fascinating stories.

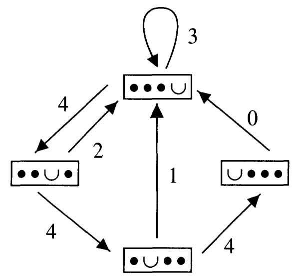

Figure 1. Juggling three balls with maximum height four. The edges are labeled with the throw height.

Remark 1. In this section, we have introduced the fundamentals of "siteswap" notation—albeit in a nontraditional form. As with many great ideas, the provenance of siteswap notation is somewhat murky. However, we can safely say that this notation was introduced (independently, and in various forms) in the mid 1980s by Paul Klimek, by Colin Wright and members of the Cambridge Juggling Club, and by Bengt Magnusson and Bruce Tiemann. For a more leisurely introduction to and a bibliography of papers about siteswap notation, the reader should consult [**4**]. General internet resources on juggling include [**1**] and [**2**]. A recent book devoted to the mathematics of juggling is [**8**].

Both as a useful notation for juggling and as a source of interesting enumerative combinatorics, it is preferable to introduce siteswap notation by defining patterns as repeating sequences of throws. Such patterns correspond to cycles in certain graphs, such as the graph of Figure 1. Many computer programs have been written to animate these patterns. See [**3**] for one cowritten by the author, or [**2**] for a comprehensive list.

To answer our random question, we need an explicit description of the set of possible states Magnus can pass through and of the possible transitions between states. To describe these transitions, we introduce a family of *throwing* operators Θ_j acting on states. Intuitively, Θ_0 corresponds to Magnus waiting when no ball lands and Θ_j corresponds to Magnus making a height j throw when a ball does land. Precisely,

$$\Theta_j(\boldsymbol{\nu}) = \begin{cases} \boxed{\nu_2 \cdots \nu_h \smile} & \text{if } j = 0, \\ \boxed{\nu_2 \cdots \nu_j \bullet \nu_{j+2} \cdots \nu_h \smile} & \text{if } j > 0. \end{cases}$$

For the remainder of this and the next section, we fix a positive maximum height h and a parameter f $(0 \le f \le h)$ that determines the number $h - f$ of balls being juggled. We are now ready to define the directed graph that specifies the possible states for Magnus, along with his possible transitions. Informally, the valid transitions correspond either to waiting or to making throws to empty landing times. The juggling state graph was apparently first considered by Jack Boyce.

Definition. The *state graph* $G_{h,f}$ is the directed graph defined as follows:

1. The vertices of $G_{h,f}$ are indexed by $St_{h,f}$.
2. There is an edge $(\boldsymbol{\nu}, \boldsymbol{\omega})$ from $\boldsymbol{\nu}$ to $\boldsymbol{\omega}$ whenever (a) $\boldsymbol{\omega} = \Theta_j(\boldsymbol{\nu})$ for some j with $1 \le j \le h$, $\nu_1 = \bullet$, and $\nu_{j+1} = \smile$ or (b) $\boldsymbol{\omega} = \Theta_0(\boldsymbol{\nu})$ and $\nu_1 = \smile$.

The set of all edges is denoted $\text{Edges}(G_{h,f})$. If $(\boldsymbol{\nu}, \boldsymbol{\omega})$ belongs to $\text{Edges}(G_{h,f})$, then we refer to $\boldsymbol{\nu}$ as a *precursor* of $\boldsymbol{\omega}$. Figure 2 illustrates $G_{5,2}$.

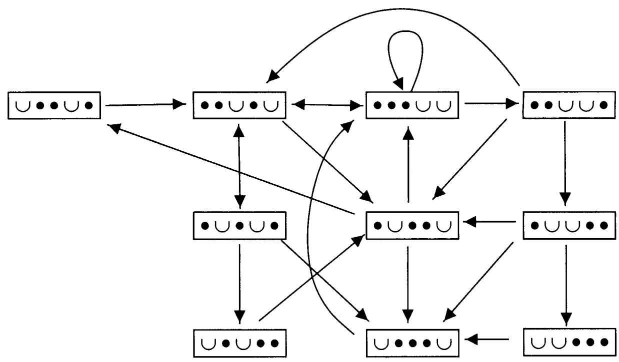

Figure 2. The state graph $G_{5,2}$.

Let us revisit the scenario introduced at the beginning of the paper (this time with Magnus juggling). When we freeze the action, Magnus is in a particular state—he is committed to making certain catches. If no ball is due to land in one second, he is forced to wait. If a ball is scheduled to land, however, he can throw to any height ($\leq h$) that will not lead to two balls landing at the same time. Our assumption in the following is that Magnus chooses, with equal probability, one of these throws. In this manner, Magnus is taking a random walk on the graph $G_{h,f}$. Having now phrased the question in a precise manner, we need only give two definitions to state the answer.

The *Stirling number of the second kind* $\left\{ {a \atop b} \right\}$ counts the number of ways of partitioning an a-element set into b blocks (irrespective of order). For example, $\left\{ {4 \atop 2} \right\} = 7$ as we can partition $\{A, B, C, D\}$ into two parts in seven ways:

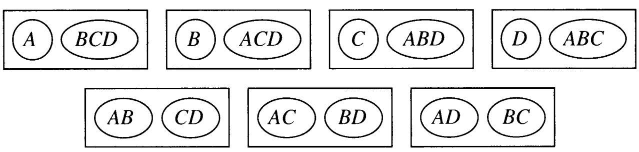

Table 1 lists $\left\{ {a \atop b} \right\}$ for a and b such that $1 \leq b \leq a \leq 6$.

Table 1. Stirling numbers of the second kind $\left\{ {a \atop b} \right\}$.

a \ b	1	2	3	4	5	6
1	1					
2	1	1				
3	1	3	1			
4	1	7	6	1		
5	1	15	25	10	1	
6	1	31	90	65	15	1

Let $\boldsymbol{\nu}$ belong to St_h. For each integer t with $1 \leq t \leq h$, we define

$$\phi_t(\boldsymbol{\nu}) = \begin{cases} |\{j : t < j \leq h, \nu_j = \smile\}| & \text{if } \nu_t = \bullet, \\ 0 & \text{if } \nu_t = \smile. \end{cases}$$

The *weight* $\Delta(\boldsymbol{\nu})$ of a state $\boldsymbol{\nu}$ is given by the formula

$$\Delta(\boldsymbol{\nu}) = \prod_{t=1}^{h} (1 + \phi_t(\boldsymbol{\nu})).$$

For example, if $\boldsymbol{\nu} = \boxed{\smile \bullet \bullet \bullet \smile \smile \smile \bullet \smile \bullet \bullet}$, then

$$\Delta(\boldsymbol{\nu}) = 1 \cdot 5 \cdot 5 \cdot 5 \cdot 1 \cdot 1 \cdot 1 \cdot 2 \cdot 1 \cdot 1 \cdot 1.$$

The weight (our terminology) arises when counting permutations with restricted positions (see, for example, [**9**, sec. 2.4]).

We are finally in a position to answer our random question.

Theorem 1. *Suppose that Magnus is juggling b balls and is able to throw them to height at most h. Then, writing f for the difference $h - b$, we can express the fraction of time Magnus spends in the state $\boldsymbol{\nu}$ of $St_{h,f}$ as*

$$\frac{\Delta(\boldsymbol{\nu})}{\left\{ {h+1 \atop f+1} \right\}}.$$

Example 1. We present here the vector $\boldsymbol{\alpha}$ whose $\boldsymbol{\nu}$th component specifies the fraction of the time Magnus spends in each state $\boldsymbol{\nu}$ of Figure 2 while juggling three balls with a maximum throw height of five.

$$\begin{aligned}
\boldsymbol{\alpha} = \frac{1}{\left\{ {6 \atop 3} \right\}} \Big(& \Delta(\boxed{\bullet \bullet \bullet \smile \smile}), \Delta(\boxed{\bullet \bullet \smile \bullet \smile}), \Delta(\boxed{\bullet \bullet \smile \smile \bullet}), \\
& \Delta(\boxed{\bullet \smile \bullet \bullet \smile}), \Delta(\boxed{\bullet \smile \bullet \smile \bullet}), \Delta(\boxed{\bullet \smile \smile \bullet \bullet}), \Delta(\boxed{\smile \bullet \bullet \bullet \smile}), \\
& \Delta(\boxed{\smile \bullet \bullet \smile \bullet}), \Delta(\boxed{\smile \bullet \smile \bullet \bullet}), \Delta(\boxed{\smile \smile \bullet \bullet \bullet}) \Big) \\
= \frac{1}{90} & (27, 18, 9, 12, 6, 3, 8, 4, 2, 1).
\end{aligned}$$

By examining $\boldsymbol{\alpha}$, we see that Magnus spends roughly one third of the time in the state corresponding to the usual three ball "cascade" juggling pattern.

Remark 2. For other contexts in which the Stirling numbers of the second kind arise in conjunction with siteswap notation, see [**5**] and [**6**]. A related class of numbers, the Eulerian numbers, also arise in the enumeration of siteswap patterns (see [**4**]).

3. PROOF OF THEOREM 1. The proof of Theorem 1 is not hard once we model the act of juggling randomly as a *Markov chain*, meaning as a discrete-time random process in which the transition probabilities depend only on the current state. To describe a Markov chain, we need to know the possible states Ω and the possible transitions between states. The latter we encode in a transition matrix P. We denote this pair by $\mathrm{MC}(\Omega, P)$. The (i, j)th entry of the matrix P is to be interpreted as the probability of transitioning from the ith state to the jth state in one step.

In describing Magnus as taking a random walk on the states $St_{h,f}$ with throws chosen uniformly randomly, we have already described one Markov chain; namely,

$\mathrm{MC}(St_{h,f}, P)$, where $P = (p_{\nu\omega})_{\nu,\omega \in St_{h,f}}$ with

$$p_{\nu\omega} = \begin{cases} 1 & \text{if } (\boldsymbol{\nu}, \boldsymbol{\omega}) \text{ is in } \mathrm{Edges}(G_{h,f}) \text{ and } \nu_1 = \smile, \\ \frac{1}{f+1} & \text{if } (\boldsymbol{\nu}, \boldsymbol{\omega}) \text{ is in } \mathrm{Edges}(G_{h,f}) \text{ and } \nu_1 = \bullet, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that we have a Markov chain with r states and transition matrix $M = (m_{ij})$ such that the lth power $M^l = (m_{ij}^{(l)})$ of M satisfies $m_{ij}^{(l)} > 0$ for l sufficiently large. This condition on the matrix M means that, no matter how long the random process has been running, it is still possible to reach any state from the current state. The process $\mathrm{MC}(St_{h,f}, P)$ satisfies this positivity condition. A *probability vector* is a vector with nonnegative entries that sum to 1. The standard theorem of Markov chains we shall use is the following [**7**, Theorem 4.1.4, p. 70]:

Theorem 2. *The following statements hold for a Markov chain with r states whose transition matrix M satisfies the stated positivity condition:*

1. $A = \lim_{l\to\infty} M^l$ *exists.*
2. *Each row of A is the same vector $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_r)$.*
3. *Each entry α_i of $\boldsymbol{\alpha}$ is positive.*
4. *The vector $\boldsymbol{\alpha}$ is the unique probability vector satisfying $\boldsymbol{\alpha} M = \boldsymbol{\alpha}$.*
5. *For any probability vector $\boldsymbol{\pi}$, $\boldsymbol{\pi} M^l \to \boldsymbol{\alpha}$ as $l \to \infty$.*

The number α_i can be interpreted as the fraction of time the Markov chain is expected to spend in the ith state. Hence, Theorem 2 tells us how to answer our question: To find the fraction of time that Magnus is in state $\boldsymbol{\omega}$, we need to calculate the $\boldsymbol{\omega}$th entry of the (normalized) left eigenvector for P with eigenvalue 1!

Example 2. We illustrate these ideas using the graph $G_{4,1}$ given in Figure 1. The transition matrix P simply consists of a $1/2$ for every edge except the unique edge leaving $\boxed{\smile\bullet\bullet\bullet}$, which must be taken. Notice that the sum of the entries in each row of P is 1. We exhibit P and two of its powers:

$$P = \begin{array}{c|cccc} & \boxed{\bullet\bullet\bullet\smile} & \boxed{\bullet\bullet\smile\bullet} & \boxed{\bullet\smile\bullet\bullet} & \boxed{\smile\bullet\bullet\bullet} \\ \hline \boxed{\bullet\bullet\bullet\smile} & 1/2 & 1/2 & & \\ \boxed{\bullet\bullet\smile\bullet} & 1/2 & & 1/2 & \\ \boxed{\bullet\smile\bullet\bullet} & 1/2 & & & 1/2 \\ \boxed{\smile\bullet\bullet\bullet} & 1 & & & \end{array}$$

$$P^2 = \begin{array}{c|cccc} & \boxed{\bullet\bullet\bullet\smile} & \boxed{\bullet\bullet\smile\bullet} & \boxed{\bullet\smile\bullet\bullet} & \boxed{\smile\bullet\bullet\bullet} \\ \hline \boxed{\bullet\bullet\bullet\smile} & 1/2 & 1/4 & 1/4 & \\ \boxed{\bullet\bullet\smile\bullet} & 1/2 & 1/4 & & 1/4 \\ \boxed{\bullet\smile\bullet\bullet} & 3/4 & 1/4 & & \\ \boxed{\smile\bullet\bullet\bullet} & 1/2 & 1/2 & & \end{array}$$

$$P^5 = \begin{array}{c|cccc} & \boxed{\bullet\bullet\bullet\smile} & \boxed{\bullet\bullet\smile\bullet} & \boxed{\bullet\smile\bullet\bullet} & \boxed{\smile\bullet\bullet\bullet} \\ \hline \boxed{\bullet\bullet\bullet\smile} & 17/32 & 9/32 & 1/8 & 1/16 \\ \boxed{\bullet\bullet\smile\bullet} & 17/32 & 1/4 & 5/32 & 1/16 \\ \boxed{\bullet\smile\bullet\bullet} & 17/32 & 1/4 & 1/8 & 3/32 \\ \boxed{\smile\bullet\bullet\bullet} & 9/16 & 1/4 & 1/8 & 1/16 \end{array}$$

Already for P^5 we see that the rows appear to be converging to the same vector. And, indeed, by Theorem 1 this vector is:

$$\begin{aligned}\boldsymbol{\alpha} &= \frac{1}{\left\{ {5 \atop 2} \right\}}\big(\Delta(\boxed{\bullet\ \bullet\ \bullet\ \smile}), \Delta(\boxed{\bullet\ \bullet\ \smile\ \bullet}), \Delta(\boxed{\bullet\ \smile\ \bullet\ \bullet}), \Delta(\boxed{\smile\ \bullet\ \bullet\ \bullet})\big) \\ &= \frac{1}{15}\,(8, 4, 2, 1)\,.\end{aligned}$$

Just over half of Magnus's time will be spent in the state $\boxed{\bullet\ \bullet\ \bullet\ \smile}$. In fact, this geometric distribution is what one would expect by examination of the graph $G_{4,1}$.

One can certainly prove Theorem 1 by guessing the vector $\boldsymbol{\alpha}$ and then proving that it satisfies the conditions of Theorem 2. However, as is often the case in mathematics, keeping track of additional information enables us to write a simpler and more illuminating proof. So, our approach will be to construct an augmented Markov chain that is easy to analyze and from which we can obtain our original Markov chain by grouping states.

We augment our states by specifying *which* balls land when. To this end, we define a *landing/throwing-state* (*LT-state*) to be a word $\widehat{\boldsymbol{\nu}} = \boxed{\widehat{\nu}_1\,\widehat{\nu}_2 \cdots\ \widehat{\nu}_h}$ of length h in the alphabet $\{\smile, 1, 2, \ldots, h\}$ such that

$$\widehat{\nu}_t \geq t \quad \text{or} \quad \widehat{\nu}_t = \smile \tag{1}$$

and

$$\widehat{\nu}_j - j \neq \widehat{\nu}_k - k \ (j \neq k,\ \widehat{\nu}_j \neq \smile,\ \widehat{\nu}_k \neq \smile). \tag{2}$$

Condition (1) ensures that any ball currently slated to land has, in fact, already been thrown. Condition (2) ensures that Magnus did not have to throw two balls at once in order to get into his current LT-state.

As is the case for states, we interpret $\widehat{\nu}_t = \smile$ to mean that no ball lands t seconds into the future. We interpret $\widehat{\nu}_t = i$ as signifying that the ball thrown $i - t$ seconds in the past will land t seconds into the future. (Alternatively, the ball landing t seconds into the future will have been in the air i seconds by the time it lands.) Let $\widehat{St}_h$ denote the set of all LT-states of length h, and let $\widehat{St}_{h,f}$ be the subset of LT-states in which $\smile$ appears exactly f times. The throwing operators Θ_j are extended in the obvious manner to act on LT-states.

Of course, each LT-state determines some landing state simply by forgetting how long each ball has been in the air. We can thus define a map $\pi_h : \widehat{St}_h \longrightarrow St_h$ by $\pi_h(\widehat{\boldsymbol{\nu}}) = \boxed{\nu_1\,\nu_2 \cdots\ \nu_h}$ for $\widehat{\boldsymbol{\nu}} = \boxed{\widehat{\nu}_1\,\widehat{\nu}_2 \cdots\ \widehat{\nu}_h}$, where

$$\nu_t = \begin{cases} \smile & \text{if } \widehat{\nu}_t = \smile, \\ \bullet & \text{if } \widehat{\nu}_t \text{ is in } \{1, 2, \ldots, h\}. \end{cases}$$

In addition to $\mathrm{MC}(St_{h,f}, P)$, we now have another natural Markov chain to consider. It is $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$, where $\widehat{P} = (\widehat{p}_{\widehat{\nu}\widehat{\omega}})_{\widehat{\nu},\widehat{\omega} \in \widehat{St}_{h,f}}$ with

$$\widehat{p}_{\widehat{\nu}\widehat{\omega}} = \begin{cases} 1 & \text{if } \widehat{\boldsymbol{\omega}} = \Theta_0(\widehat{\boldsymbol{\nu}}) \text{ and } \widehat{\nu}_1 = \smile, \\ \frac{1}{f+1} & \text{if } \widehat{\boldsymbol{\omega}} = \Theta_j(\widehat{\boldsymbol{\nu}}) \text{ for some } j,\ \widehat{\nu}_1 \neq \smile, \text{ and } \widehat{\nu}_{j+1} = \smile, \\ 0 & \text{otherwise.} \end{cases}$$

Our approach to proving Theorem 1 will be to analyze this latter Markov chain. The first step is to find the vector $\boldsymbol{\beta}$ of steady-state probabilities for $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$ (i.e., the vector $\boldsymbol{\beta}$ that appears as each row of $B = \lim_{l\to\infty} \widehat{P}^l$). The second step is to show that by appropriately "lumping" the states of this chain we recover our original chain $\mathrm{MC}(St_{h,f}, P)$. By analyzing this lumping process carefully, it will become clear how to obtain the result of Theorem 1 from our knowledge of the vector $\boldsymbol{\beta}$.

Lemma 1. *The vector* $(1, 1, \dots, 1)$ *(of length* $|\widehat{St}_{h,f}|$*) is a left eigenvector for* $\widehat{P}$.

Proof. To prove this, we must simply show that the transition matrix $\widehat{P}$ is doubly stochastic (i.e., that the rows *and* columns of $\widehat{P}$ sum to 1). There are two cases to consider.

In the first case, one of $\widehat{\nu}_t = \smile$ or $\widehat{\nu}_t > t$ holds for each t with $1 \le t \le h$. In this scenario, Magnus has waited during the most recent beat. That means that the only precursor of $\widehat{\boldsymbol{\nu}}$ is $\boxed{\smile \widehat{\nu}_1 \cdots \widehat{\nu}_{h-1}}$. The alternative is that $\widehat{\nu}_t = t$ holds for some t, in which event he has just made a height t throw. Since Magnus throws at most one ball at any given time, there is at most one such t. By the discussion in the proof of Lemma 4, the previous arc of this ball could have been any of $f + 1$ different heights. Each of these different heights leads to a different precursor. Since the out-degree of any state $\widehat{\boldsymbol{\nu}}$ with $\widehat{\nu}_1 \neq \smile$ is $f + 1$, each of these edges has weight $1/(f + 1)$. This proves that the columns of $\widehat{P}$ also sum to 1. ∎

In order to describe fully the vector $\boldsymbol{\beta}$ for $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$, we need to calculate $|\widehat{St}_{h,f}|$.

Lemma 2. *The set* $\widehat{St}_{h,f}$ *satisfies* $|\widehat{St}_{h,f}| = \left\{ {h+1 \atop f+1} \right\}$.

Proof. We proceed by constructing a bijection between elements of $\widehat{St}_{h,f}$ and partitions with $f + 1$ blocks of the set $[h + 1] := \{1, 2, \dots, h + 1\}$. An example of the correspondence we construct is illustrated in Figure 3.

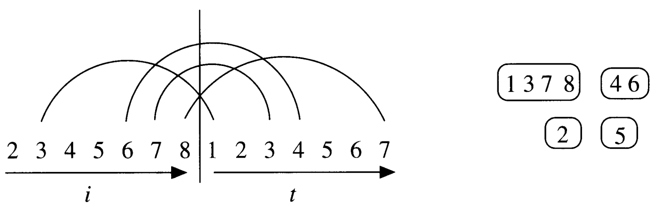

Figure 3. The arcs on the left correspond to the edges of the graph $\Gamma_{\widehat{\nu}}$ for the LT-state $\widehat{\boldsymbol{\nu}} = \boxed{6 \smile 4\,6 \smile\smile 7}$. On the right is the corresponding partition $\lambda(\widehat{\boldsymbol{\nu}})$. Notice that $f = 3$ and the partition on the right has four blocks.

We start by assigning an undirected graph $\Gamma_{\widehat{\nu}}$ with vertex set $[h + 1]$ to each $\widehat{\boldsymbol{\nu}}$ in $\widehat{St}_{h,f}$. When $1 \le t < i \le h + 1$, we join t and i by an edge $\{t, i\}$ of $\Gamma_{\widehat{\nu}}$ if and only if there was a ball thrown $h + 1 - i$ seconds in the past that will land t seconds into the future. In terms of $\widehat{\boldsymbol{\nu}}$, this means simply that $\widehat{\nu}_i = h + 1 + t - i$.

Let $\lambda(\widehat{\boldsymbol{\nu}})$ be the set partition of $[h + 1]$ corresponding to the connected components of $\Gamma_{\widehat{\nu}}$. Our goal is to show that the map $\widehat{\boldsymbol{\nu}} \mapsto \lambda(\widehat{\boldsymbol{\nu}})$ is a bijection onto partitions of $[h + 1]$ with $f + 1$ blocks.

To begin, we examine the connected components of $\Gamma_{\widehat{\nu}}$. Suppose that $\{a, b\}$ and $\{a, c\}$ are distinct edges of $\Gamma_{\widehat{\nu}}$. It cannot be true that both $a < b$ and $a < c$, for if that were the case Magnus would be planning to catch two balls $h+1-a$ seconds from now. Nor can it be true that both $b < a$ and $c < a$, as this would imply that a seconds ago Magnus threw two balls at once. We conclude that either $b < a < c$ or $c < a < b$. It follows that the deletion of any edge from $\Gamma_{\widehat{\nu}}$ increases the number of connected components by one.

The number of edges in $\Gamma_{\widehat{\nu}}$ is $h - f$ since there is an edge for each j with $\widehat{\nu}_j \neq \smile$. So, by the previous paragraph, the number of connected components in $\Gamma_{\widehat{\nu}}$ is the number of vertices (i.e., $h+1$) decreased by the number of edges (i.e., $h-f$). This yields $f+1$ connected components. Hence, $\lambda(\widehat{\boldsymbol{\nu}})$ is a partition of $[h+1]$ into $f+1$ blocks as desired. Furthermore, it is easily checked that the block structure of $\lambda(\widehat{\boldsymbol{\nu}})$ uniquely determines the LT-state $\widehat{\boldsymbol{\nu}}$. Thus, the map $\widehat{\boldsymbol{\nu}} \mapsto \lambda(\widehat{\boldsymbol{\nu}})$ is an injective map from $\widehat{St}_{h,f}$ to partitions of $[h+1]$ with $f+1$ blocks.

Now suppose that λ is a (set) partition of $[h+1]$. We associate an LT-state $\widehat{\boldsymbol{\omega}}(\lambda) = \boxed{\widehat{\omega}(\lambda)_1 \cdots \widehat{\omega}(\lambda)_h}$ with λ as follows. Consider a block $\{\alpha_1, \alpha_2, \ldots, \alpha_m\}$ (indexed such that $\alpha_i < \alpha_j$ if $i < j$) of λ. If $m = 1$, then set $\widehat{\omega}(\lambda)_{\alpha_1} = \smile$. Otherwise, set $\widehat{\omega}(\lambda)_{\alpha_l} = h + 1 + \alpha_l - \alpha_{l+1}$ for each l with $1 \leq l < m$. Since $1 \leq \alpha_l < \alpha_{l+1} \leq h+1$ for each l, $\widehat{\boldsymbol{\omega}}(\lambda)$ is certainly a word of length h in the alphabet $\{\smile, 1, 2, \ldots, h\}$. That (1) is satisfied follows from the inequalities $\alpha_l < \alpha_{l+1} \leq h+1$. That $\widehat{\boldsymbol{\omega}}(\lambda)$ satisfies (2) is a consequence of the fact that the blocks in λ are disjoint. Hence, $\widehat{\boldsymbol{\omega}}(\lambda)$ is, in fact, an LT-state. Therefore, every partition of $[h+1]$ into $f+1$ blocks is obtained as a $\lambda(\widehat{\boldsymbol{\nu}})$ for a unique $\widehat{\boldsymbol{\nu}}$ from $\widehat{St}_{h,f}$. This completes the proof. ■

Corollary 1. *In the process* $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$*, the fraction of time that Magnus finds himself in a state* $\widehat{\boldsymbol{\nu}}$ *is* $1/\left\{ {h+1 \atop f+1} \right\}$.

We have shown that, when juggling randomly, Magnus is just as likely to find himself in one LT-state as in another. We now consider a new random process running in parallel with $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$. For $\boldsymbol{\nu}$ in $St_{h,f}$, we define $[\boldsymbol{\nu}] = \pi_h^{-1}(\boldsymbol{\nu})$, which is a subset of $\widehat{St}_{h,f}$. When $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$ is in state $\widehat{\boldsymbol{\nu}}$, we define our new process to be in the state $[\pi_h(\widehat{\boldsymbol{\nu}})]$. It follows that its states are $\{[\boldsymbol{\nu}]\}_{\boldsymbol{\nu} \in St_{h,f}}$. This is a *lumped* process derived from $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P})$ by grouping together certain states (see Figure 4).

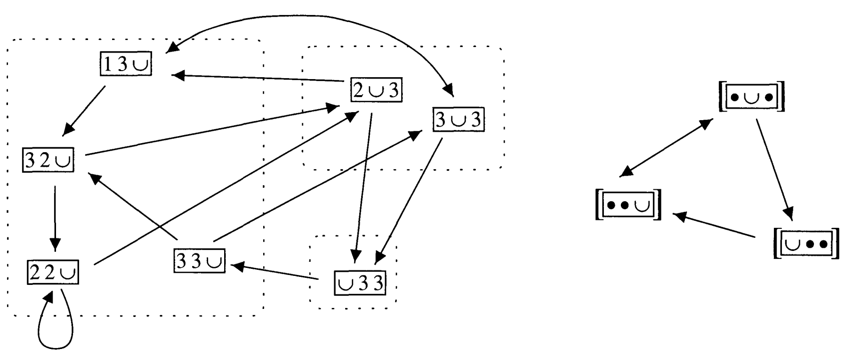

Figure 4. The Markov chain $\mathrm{MC}(\widehat{St}_{3,1}, \widehat{P})$ along with the corresponding lumped process.

Certainly Magnus wanders randomly among the states of this new process. We do not yet know, however, that this new process is a Markov chain. In particular, it is not clear that the probability of transitioning from $[\boldsymbol{\nu}]$ to $[\boldsymbol{\omega}]$ is independent of how long Magnus has been in $[\boldsymbol{\nu}]$. If it *were* dependent, then the transition probabilities would depend on previous states as well as the current state. Or, stated another way, the transition probability from $[\boldsymbol{\nu}]$ to $[\boldsymbol{\omega}]$ would depend on *which* state $\widehat{\boldsymbol{\mu}}$ of $[\boldsymbol{\nu}]$ Magnus is currently in.

For each $\widehat{\boldsymbol{\mu}}$ in $[\boldsymbol{\nu}]$, let $\widehat{p}_{\widehat{\mu},[\omega]}$ denote the probability of $\mathrm{MC}(\widehat{St}_{h,f}, \widehat{P},)$ transitioning from $\widehat{\boldsymbol{\mu}}$ to some element of $[\boldsymbol{\omega}]$. The foregoing discussion suggests (see [**7**, Theorem 6.3.2, p. 124] for a proof) that our new process will be a Markov chain if the probabilities $p_{\widehat{\mu},[\omega]}$ are equal for each $\widehat{\boldsymbol{\mu}}$ in $[\boldsymbol{\nu}]$. Furthermore, the transition probabilities $r_{[\nu][\omega]}$ of our new process will be these common values $\widehat{p}_{\widehat{\mu},[\omega]}$.

Lemma 3. *If $\boldsymbol{\nu}$ and $\boldsymbol{\omega}$ belong to $St_{h,f}$, then $\widehat{p}_{\widehat{\mu},[\omega]} = p_{\nu\omega}$ for each $\widehat{\boldsymbol{\mu}}$ in $[\boldsymbol{\nu}]$.*

Proof. If $p_{\nu\omega} = 0$, then certainly all of the $\widehat{p}_{\widehat{\mu},[\omega]}$ are 0 as well. Also, if $p_{\nu\omega} = 1$, then Magnus must wait in order to transition from any state $\widehat{\boldsymbol{\mu}}$ in $[\boldsymbol{\nu}]$, whence $\widehat{p}_{\widehat{\mu},[\omega]} = 1$. So the only case to consider is when $p_{\nu\omega} = 1/(f+1)$. In this case $\boldsymbol{\omega} = \Theta_j(\boldsymbol{\nu})$ for some j with $1 \le j \le h$, $\nu_1 = \bullet$, $\nu_{j+1} = \smile$, and $\omega_j = \bullet$. Suppose that Magnus transitions from $\widehat{\boldsymbol{\mu}}$ to a particular $\widehat{\boldsymbol{\omega}}$ in $[\boldsymbol{\omega}]$. The previous equalities imply that $\widehat{\mu}_{j+1} = \smile$ and $\widehat{\omega}_j = j$. It follows that $\widehat{\boldsymbol{\omega}}$ is obtained from $\widehat{\boldsymbol{\mu}}$ by making a height j throw. This will be one of $f+1$ possible throws for Magnus because $\widehat{\mu}_1 \ne \smile$, so he will make this choice with probability $1/(f+1)$. As we have considered all cases, this completes the proof of the lemma. ∎

Define a matrix $R = (r_{[\nu][\omega]})$, with rows and columns indexed by $\{[\boldsymbol{\mu}]\}_{\mu\in St_{h,f}}$, by setting $r_{[\nu][\omega]} = p_{\nu\omega}$. In view of Lemma 3, we have obtained a new stochastic process $\mathrm{MC}(\{[\boldsymbol{\mu}]\}_{\mu\in St_{h,f}}, R)$ that can be identified in the obvious manner with $\mathrm{MC}(St_{h,f}, P)$. It follows from Corollary 1 that the fraction of the time Magnus finds himself in a state $\boldsymbol{\nu}$ with respect to the process $\mathrm{MC}(St_{h,f}, P)$ is $|[\boldsymbol{\nu}]|/\left\{{h+1 \atop f+1}\right\}$. Hence, Theorem 1 follows from Lemma 4.

Lemma 4. *For a state $[\boldsymbol{\nu}]$ from* $\mathrm{MC}(\{[\boldsymbol{\nu}]\}_{\nu\in St_{h,f}}, R)$,

$$|[\boldsymbol{\nu}]| = \Delta(\boldsymbol{\nu}) = \prod_{t=1}^{h} (1 + \phi_t(\boldsymbol{\nu})).$$

Proof. Let $\widehat{\boldsymbol{\nu}}$ in $\widehat{St}_{h,f}$ and $\boldsymbol{\nu}$ in $St_{h,f}$ be such that $\pi_h(\widehat{\boldsymbol{\nu}}) = \boldsymbol{\nu}$. Suppose that $\nu_t = \bullet$ and that we know the value $\widehat{\nu}_i$ whenever $i > t$. How do these values constrain the possibilities for $\widehat{\nu}_t$? To start, note that the ball landing t seconds from now was thrown at most $h-t$ seconds in the past. As Magnus must have already thrown this ball, there are at most $h-t+1$ possibilities for $\widehat{\nu}_t$. In addition, any ball landing *more* than t seconds into the future must have been thrown *fewer* than $h-t$ seconds in the past, for $\boldsymbol{\nu}$ belongs to St_h. Combining these two observations, we see that the number of possibilities for $\widehat{\nu}_t$ is reduced precisely by the number of balls landing after it. In particular, the possible values for $\widehat{\nu}_t$ depend only on the ν_i with $i > t$.

In conclusion, the number of possible values for $\widehat{\nu}_t$ when $\nu_t = \bullet$ is

$$h - t + 1 - |\{i : i > t \text{ and } \nu_i = \bullet\}|,$$

which is just $1 + \phi_t(\boldsymbol{\nu})$. Taking the product over all t yields the desired answer. ∎

This completes the proof of Theorem 1.

Remark 3. Consideration of the powers of the transition matrix P might lead to new asymptotic formulas for the Stirling numbers of the second kind. Consider $G_{h,f}$, the associated transition matrix P, and the state $\boldsymbol{\omega} = \boxed{\smile \cdots \smile \bullet \cdots \bullet}$ from $St_{h,f}$. We have already seen that the rows of P^l converge to a probability vector $\boldsymbol{\alpha}$ with $\alpha_\omega = 1/\left\{ {h+1 \atop f+1} \right\}$. Hence, for each state $\boldsymbol{\mu}$ in $St_{h,f}$, we get a sequence $(p_{\mu\omega}, p^{(2)}_{\mu\omega}, \dots)$ that converges to $1/\left\{ {h+1 \atop f+1} \right\}$.

4. OTHER MODELS. By now (unless you are a very fast reader), Magnus is getting tired. He is going to drop the occasional ball. Certainly any realistic juggling model should accommodate transitions to states with fewer balls. Of course, if we allow Magnus to juggle indefinitely, but he drops balls occasionally, he will eventually run out of balls. To keep things interesting, we will give his assistant Sphagnum the option of inserting a ball into Magnus's pattern whenever Magnus has a wait.

We explore two such models in this section. In both models, Magnus can simply fail to catch a ball. When this happens, he transitions from a state in $St_{h,f}$ to one in $St_{h,f+1}$. The difference between the two models lies in what is considered to be a legal throw. In the first model, Magnus (or his assistant) throws to heights that would not lead to two balls landing at the same time. In the second model, however, we allow for this eventuality. Of course, as Magnus is well known to have small hands, he cannot catch two balls at once, so he will have to ignore one of the balls. In terms of transitions, such a throw is treated no differently than a drop.

These changes are easily incorporated into the framework that we have developed. We now define the state graphs corresponding to the aforementioned *add-drop model* (a concept well known to college students) and *annihilation model.*

1. *Add-drop juggling* generalizes standard juggling by always allowing throws of height 0 and removing the restriction on $\nu_1 = \bullet$ for throws of positive height. (The throws of height 0 correspond to Magnus dropping or waiting; those of positive height when $\nu_1 = \smile$ correspond to Sphagnum helping.) So, our graph G_h^{a-d} has (a) a vertex for each state in St_h and (b) an edge $(\boldsymbol{\nu}, \boldsymbol{\omega})$ from $\boldsymbol{\nu}$ to $\boldsymbol{\omega}$ whenever (i) $\boldsymbol{\omega} = \Theta_j(\boldsymbol{\nu})$ for some j with $1 \le j \le h$ such that $\nu_{j+1} = \smile$ or (ii) $\boldsymbol{\omega} = \Theta_0(\boldsymbol{\nu})$. Figure 5 illustrates G_3^{a-d}.

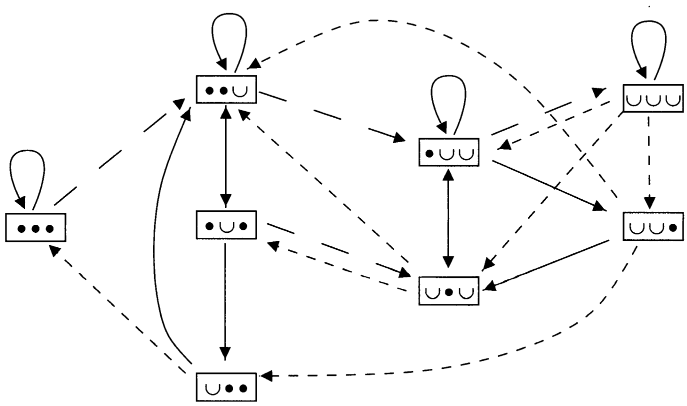

Figure 5. The state graph G_3^{a-d}. Solid arrows are regular throws; dashed are drops; dotted are insertions by Sphagnum.

2. In *annihilation juggling*, we take the same state graph $G_h^{ann} = G_h^{a-d}$ as in the previous case. The difference lies in the way we assign probabilities to the edges.

With our state graphs now defined, we can define the corresponding transition matrices Q and R. For $\boldsymbol{\nu}$ in $St_{h,f}$, set $f'_{\nu} = |\{t : 2 \le t \le h \text{ and } \nu_t = \smile\}|$.

1. For G_h^{a-d}, we take as the transition matrix $Q = (q_{\nu\omega})_{\nu,\omega \in St_h}$ with

$$q_{\nu\omega} = \begin{cases} \frac{1}{f'_{\nu}+2} & \text{if } (\boldsymbol{\nu}, \boldsymbol{\omega}) \text{ is in } \operatorname{Edges}(G_h^{a-d}), \\ 0 & \text{otherwise.} \end{cases}$$

The denominator comes from the fact that Magnus can either drop/wait or throw to any $\smile$ occurring at ν_2 or later. As he does in the standard model, Magnus chooses uniformly from the available edges.

2. For G_h^{ann}, we use the transition matrix $R = (r_{\nu\omega})_{\nu,\omega \in St_h}$ with

$$r_{\nu\omega} = \begin{cases} \dfrac{h - f'_{\nu}}{h+1} & \text{if } (\boldsymbol{\nu}, \boldsymbol{\omega}) \text{ is in } \operatorname{Edges}(G_h^{ann}) \text{ and } \Theta_0(\boldsymbol{\nu}) = \boldsymbol{\omega}, \\ \dfrac{1}{h+1} & \text{if } (\boldsymbol{\nu}, \boldsymbol{\omega}) \text{ is in } \operatorname{Edges}(G_h^{ann}) \text{ and } \Theta_0(\boldsymbol{\nu}) \neq \boldsymbol{\omega}, \\ 0 & \text{otherwise.} \end{cases}$$

Here, the reader should envision Magnus (possibly with the help of Sphagnum) throwing to *any* height ($\le h$) at random, regardless of whether or not a ball has just landed. If there is already a ball landing at the time for which he is aiming, the latter ball gets annihilated.

In order to describe the steady-state probabilities for the add-drop model, we must introduce the Bell numbers. The hth *Bell number* B_h is defined by $B_h = \sum_{i=0}^{h} \left\{ {h \atop i} \right\}$. For these two models, the analogue of Theorem 1 is:

Theorem 3. *Let $\boldsymbol{\nu}$ be a state from $St_{h,f}$.*

1. *In the add-drop model, Magnus spends the fraction*

$$\frac{\Delta(\boldsymbol{\nu})}{B_{h+1}}$$

of his time in state $\boldsymbol{\nu}$.

2. *In the annihilation model, Magnus spends the fraction*

$$\frac{\frac{h!}{(h-f)!}\Delta(\boldsymbol{\nu})}{(h+1)^h}$$

of his time in state $\boldsymbol{\nu}$.

As the denominators for our probabilities in the standard and add-drop models have natural combinatorial interpretations, the reader may be wondering if such an interpretation exists for the $(h+1)^h$ occurring in the annihilation model. Indeed, $(h+1)^h$ counts the number of labeled rooted trees with h nodes (see [**10**, Proposition 5.3.2, p. 25] for details).

Example 3. Here is the transition matrix for G_3^{a-d}:

	• • •	• • ◡	• ◡ •	◡ • •	• ◡ ◡	◡ • ◡	◡ ◡ •	◡ ◡ ◡
• • •	1/2	1/2						
• • ◡		1/3	1/3		1/3			
• ◡ •		1/3		1/3		1/3		
◡ • •	1/2	1/2						
• ◡ ◡					1/4	1/4	1/4	1/4
◡ • ◡		1/3	1/3		1/3			
◡ ◡ •		1/3		1/3		1/3		
◡ ◡ ◡					1/4	1/4	1/4	1/4

From Theorem 3(1), we obtain the probability vector

$$\boldsymbol{\alpha} = \frac{1}{15}\,(1, 4, 2, 1, 3, 2, 1, 1)\,.$$

This gives the fraction of the time Magnus spends in each of the eight possible states (the ordering of $\boldsymbol{\alpha}$ is the same as that of the transition matrix).

As the steps in the proof of Theorem 3 are analogous to those found in the proof of Theorem 1, we do not elaborate on them here. As a consolation to the reader, though, we mention two other families of juggling models that, although perhaps less physically realistic than the ones we have described, do lead to interesting answers to our random question.

1. *Multiplex juggling*. It is not necessary (assuming you find a juggler with bigger hands) to disallow the catching and throwing of more than one ball at a time. The standard and add-drop models can both be reinterpreted with this condition relaxed.
2. *Infinite juggling*. Given a juggler in better shape than Magnus, it is not necessary to place a maximum on legal throw heights. In the standard juggling model, placing a probability of $(p-1)/p^j$ for the throw to the jth next available height yields tractable calculations for any p larger than 1.

   Rather than choosing geometric weights for the throws, one could choose probabilities from any convergent infinite series. For example, why not choose $6/(\pi^2 j^2)$ for the jth available throw height? Or even $90/(\pi^4 j^4)$? The sky's the limit.

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A Very Short Proof of Cauchy's Interlace Theorem

We use an overlooked characterization of interlacing to give a two-sentence proof of Cauchy's interlace theorem [**2**]. Recall that if polynomials $f(x)$ and $g(x)$ have all real roots $r_1 \le r_2 \le \cdots \le r_n$ and $s_1 \le s_2 \le \cdots \le s_{n-1}$, respectively, then these roots *interlace* if

$$r_1 \le s_1 \le r_2 \le s_2 \le \cdots \le s_{n-1} \le r_n.$$

The following can be found in [**1**], along with a discussion of its history going back to Hermite.

Theorem 1. *The roots of polynomials f and g interlace if and only if the linear combination $f + \alpha g$ has all real roots for each α in $\mathbb{R}$.*

Corollary 1. *If A is a Hermitian matrix and B is a principle submatrix of A, then the eigenvalues of B interlace the eigenvalues of A.*

Proof. Choose α in $\mathbb{R}$, partition $A = \left(\begin{array}{c|c} B & c \\ \hline c^* & d \end{array}\right)$, and consider the following equation that follows from the linearity of the determinant:

$$\left|\begin{array}{c|c} B - xI & c \\ \hline c^* & d - x + \alpha \end{array}\right| = \left|\begin{array}{c|c} B - xI & c \\ \hline c^* & d - x \end{array}\right| + \left|\begin{array}{c|c} B - xI & c \\ \hline 0 & \alpha \end{array}\right|$$

Since the determinant on the left-hand side is the characteristic polynomial of a Hermitian matrix, $|A - xI| + \alpha|B - xI|$ has all real roots for any α, whence the eigenvalues in question interlace.

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——Submitted by Steve Fisk, Bowdoin College